

The Effects of Voids on Solder Joint Reliability in First Level Interconnect

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ABSTRACT

First level interconnect (FLI), e.g. flip-chip attach on substrate or wafer, using solder as the interconnect material, is employed extensively in the advance packaging assembly. During the flip-chip die attach process, voids are often formed in the solder joints. In the past this was not seen as a major issue due to void % being typically lower than 10%, but as miniaturization continues, the solder joint of FLI is becoming smaller (<100um bump or Cu pillar size) with tighter pitch and higher density. This has resulted in an increasing concern on how the micro voids in the solder joint will affect the package's long-term reliability, despite the void % being low. In addition, a literature search on the web shows that there is limited research done on this subject.

During the first phase of this project, process recipes were defined to consistently build packages both with and without solder voids. The appropriate X-ray inspection tools and CT were used to determine the size and location of the macro and micro voids in the FLI solder joints. It was found that packages built using solder paste typically had more voids in the solder joints when compared to packages built using flux and solder sphere process. Subsequently, test packages built in the first phase, both with and without micro voids, were subjected to different reliability tests, i.e. thermal cycling tests, electromigration tests and thermal shock tests. Observations from the test results showed that test packages with solder voids generally performed worse in solder joint reliability when compared to the test packages without solder voids. It was also observed that voids can create current crowding in the area surrounding them, which accelerates electromigration degradation, with larger voids having more significant impact from the simulation model. Consistent with other studies, greater electromigration degradation occurred in joints where the orientation of Sn grains are aligned along the c-axis and parallel to the current flow direction.

This paper will describe the details of the different test conditions, and discuss observations from the failure analysis done with cross-sections and EBSD analysis.

INTRODUCTION

With the emergence of advanced packaging technology and heterogeneous integration, micro Cu-pillar bump is widely used as FLI. Micro-voids are often found in this kind of FLI flip-chip attach and there are limited insights on the effects of voids on the reliability of the flip-chip solder joint. A project team was formed under The International Electronics Manufacturing Initiative (iNEMI) Packaging TIG to study the impact of micro-voids on electromigration, thermal cycling, and thermal shock performance.

In the first phase of this project [1], process recipes that consistently created test packages with and without solder voids were defined (Figure 1). Lead-free solder, Sn-3.0Ag-0.5Cu (SAC305) was chosen as the solder alloy. A void inspection capability study was carried out using X-ray and computed tomography to determine the level of voiding in the test packages. In the second phase, the test

packages were subjected to three different reliability tests. Electromigration (EM) was used to assess the gradual migration of material in high current density structures, such as in the first-level interconnects. Severe electromigration can result in interconnect cavitation which will eventually lead to circuit failure. Thermal cycle and thermal shock were used to assess the effect of thermal expansion mismatch between the silicon die and the substrate. Differing material expansion can create high solder joint stresses at die corners and around the die edges. Voids in the interconnects could potentially become structural flaws that might lead to early joint fracture. A comparison was made to assess the solder joint reliability of test packages with and without solder voids under the said reliability test conditions.

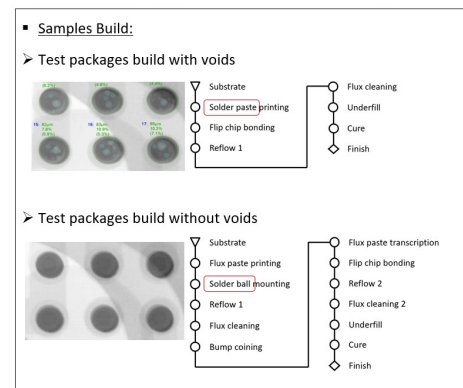


FIGURE 1. PROCESS FLOW FOR TEST VEHICLE BUILT

EXPERIMENT

Electromigration (EM)

Test packages (Figure 2) were subjected to an electromigration test according to the JEDEC standard JEP154 [2]. The test was conducted at a temperature of 150°C with current density of 40kA/cm² applied across the solder resist opening area. Figure 2 also shows the cross-section view of an interconnect. Both forward and reverse current flow directions were assessed in the test. A 10% resistance increase was set as the joint failure criterion. Tested packages were sent for void inspection using X-ray, and a selected number of units were sent for cross-section analysis using Scanning Electron Microscopy (SEM) and Electron Backscatter Diffraction (EBSD).

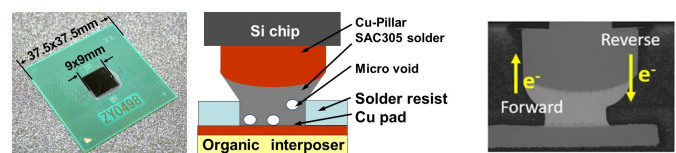


FIGURE 2. ELECTROMIGRATION TEST PACKAGE

Thermal cycling and thermal shock

Slightly different test packages with similar package sizes were subjected to thermal cycle and thermal shock tests in accordance to JESD22-A104D and JESD22-A106B. The upper and lower temperature range of 125°C and -55°C were chosen for both thermal cycle and thermal shock tests. Another test condition of -65°C to 150°C was also added for thermal shock test due to the chamber availability later in the experiment. Tested packages were sent for void inspection using X-ray and a few units were sent for cross-section analysis using SEM.

RESULTS AND DISCUSSION

Electromigration (EM)

Figures 3 and 4 show the test results for forward and reverse current flow directions respectively. It was observed that the presence of voids reduced the average lifetime of test packages by slightly more than half. The lifetimes in the reverse direction were approximately 4.3 times longer than those in the forward direction.

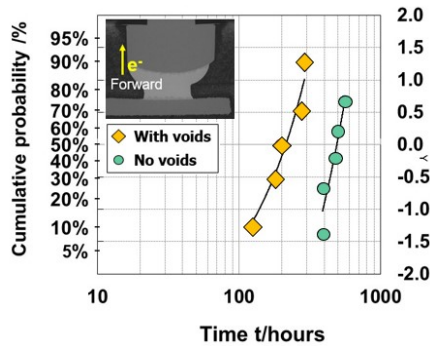


FIGURE 3. CUMULATIVE PROBABILITY OF FAILURE FOR FORWARD CURRENT FLOW DIRECTION

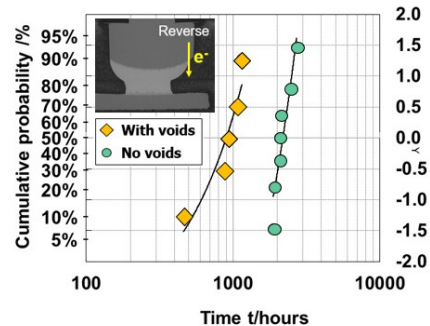


FIGURE 4. CUMULATIVE PROBABILITY OF FAILURE FOR REVERSE CURRENT FLOW DIRECTION

SEM of typical bumps built with paste (with micro-voids), in the early stage of EM test are shown in Figure 5. Micro voids were observed at the Cu pillar/solder interface and at the solder/Cu pad interface. Significant growth of intermetallic compounds (IMCs) was observed near the micro-voids.

[3] Figure 6 shows another example of a Cu pillar bump after EM testing for 8 hours. The reddish color shows that the c-axis [001] of the β -Sn grain is oriented in the current direction. When this happened, large diffusion of β -Sn and formation of IMCs were observed. The presence of micro voids is expected to accelerate formation of IMCs.

Another joint that has undergone 8hrs of EM is shown in Figure 7. The presence of a void has created current crowding and current shadowing around its edges. IMC growth is driven by the current

flow mainly at the edges of the void, so IMC growth is greatest in those areas. Moreover, Sn regions with (001) grain orientation (in red color) have the highest diffusivities. This aligns with the known relation between electromigration anisotropy and Sn crystal structure orientation [4]. Figure 7 shows a higher diffusion rate at the right side of the joint where the EBSD image quality map confirmed that the c-axis of β -Sn crystal structure was aligned with the direction of the current flow.

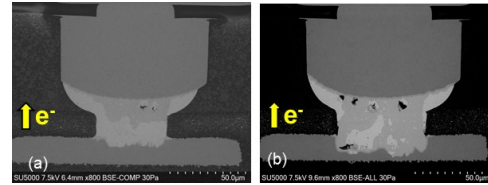


FIGURE 5. CROSS-SECTION ON TYPICAL BUMPS OF PASTE SAMPLE (A) AFTER 8HRS OF EM TEST, (B) AFTER 48HRS OF EM TEST

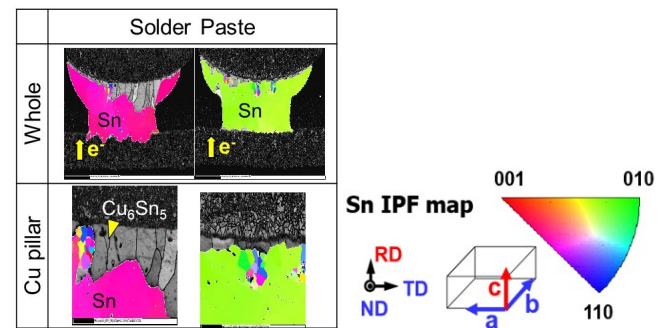


FIGURE 6. EBSD IMAGE QUALITY MAP AND INVERSE POLE FIGURE OF β -Sn IN THE CU PILLAR BUMP AFTER EM TEST FOR 8 HOURS.

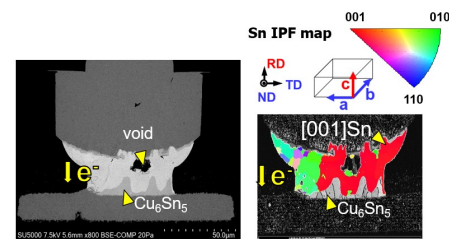


FIGURE 7. CROSS-SECTION AND EBSD IMAGE QUALITY MAP OF AN INTERCONNECT WITH SOLDER VOID, AFTER 8HRS EM TEST

Figures 8 and 9 are the cross-section and EBSD image quality maps of solder joints that went through 123hrs and 290hrs of EM testing. The joint on the left is a reference joint without any current flow, whereas the center and right joints are 2 joints from the same daisy chain. Again, higher Cu diffusion was observed when the c-axis of β -Sn crystal structure is aligned to current flow direction (red color). Green and blue color zones in the EBSD images indicate crystal orientations (010) and (110) that provide a less favorable path for electrons that drive electromigration.

A simulation model was constructed to study the current distribution of the flip-chip solder joint. The distribution of current density in the entire bump is shown in Figures 10 to 12. In the forward direction (Figure 11), the maximum current density of 42.0k A/cm² occurred at the edge of the Cu pad. In the case of the reverse direction (Figure 12), maximum current density of 21.3kA/cm² occurred at the center of the solder bump. This clearly explains the results observed in Figures 3 and 4, that the EM lifetime in the reverse current direction was approximately 4 times longer than in the forward direction.

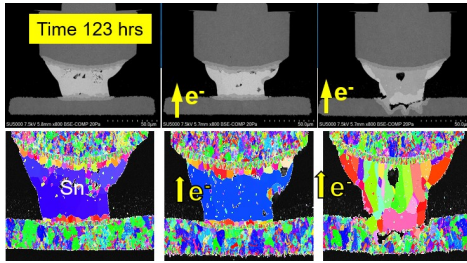


FIGURE 8. CROSS-SECTION AND EBSD IMAGE QUALITY MAP OF JOINTS, AFTER 123HRS OF EM TEST

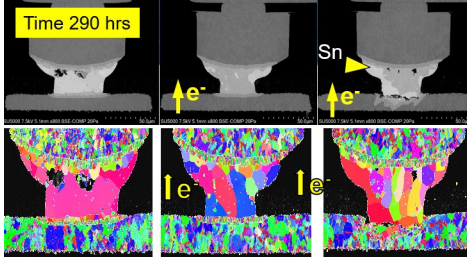


FIGURE 9. CROSS-SECTION AND EBSD IMAGE QUALITY MAP OF JOINTS, AFTER 123HRS OF EM TEST

Figure 13 shows that according to the simulation model, there was an increase in current concentration around voids. Larger voids had higher impact on current crowding. Figure 14 summarizes the results of the MTTF (hrs) for EM in comparison to the average void area prior to EM. With slightly more voids generated by the paste process, the test samples failed earlier than the test samples prepared by the ball/flux process, with little to no voiding. It is also clear that the MTTF for forward current direction is shorter than the reverse current direction. Please note that failure in EM was also impacted by Sn grain structure orientation as discussed earlier. These results agree with the simulation model, and with observations from the cross-section and EBSD analysis.

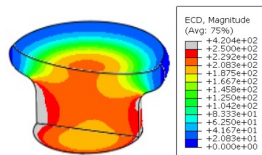


FIGURE 10. CURRENT DENSITY DISTRIBUTION FOR WHOLE BUMP

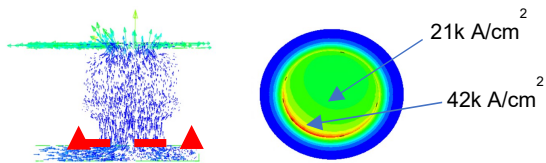


FIGURE 11. FORWARD ELECTRON FLOW IMAGE AND CURRENT DENSITY DISTRIBUTION AT SOLDER AND CU PAD INTERFACE

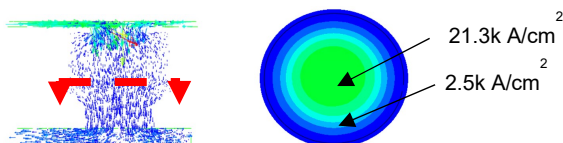


FIGURE 12. REVERSE ELECTRON FLOW IMAGE AND CURRENT DENSITY DISTRIBUTION AT SOLDER AND CU PILLAR INTERFACE

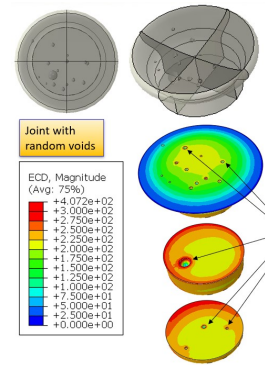


FIGURE 13. CURRENT DENSITY DISTRIBUTION AROUND VOID

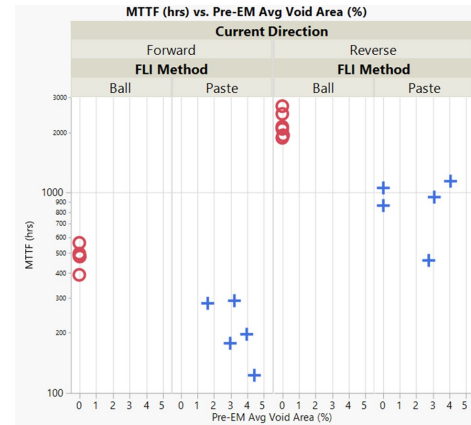


FIGURE 14. MTTF (HRS) OF EM VS PRE-EM AVERAGE VOID

Thermal cycling (TC)

Results of thermal cycling are summarized in Figure 15. There was no failure from the test packages that were built using the ball/flux process with minimal voiding, even up to 2000 cycles. 2 test packages that were built using the paste process, with voids ranging from 6-9% pre-TC, observed high resistance at 2000 cycles. One of these test packages was further analyzed with 3D X-Ray and cross-section (Figure 16). There was no anomaly noticed from the 3D X-Ray. The only observation was void area % had reduced to less than 5% post-TC. From the cross-section in Figure 16, the underfill fillet height was significantly lower at this corner compared to the rest. It is likely that the thermal mechanical stress at the die corner during TC, the delamination along die to underfill interface, initiated the crack that propagated to the Cu-passivation interface and finally cracked through the Cu bump, as shown by the blue arrows.

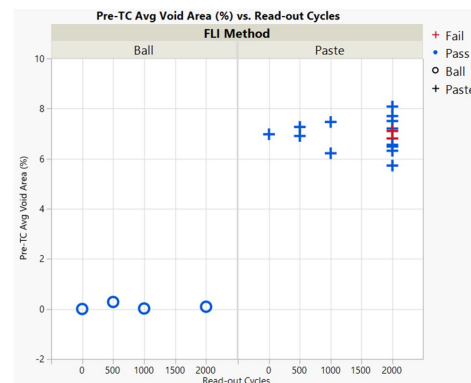


FIGURE 15. PRE-TC AVERAGE VOID VS TC FAILURE CYCLE COUNT

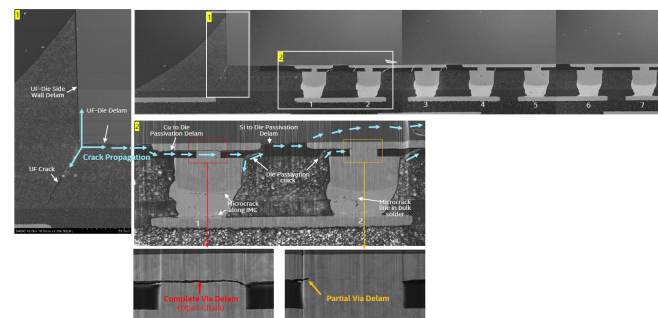


FIGURE 16. CROSS-SECTION OF HIGH RESISTANCE TEST SAMPLE POST 2000 CYCLES OF TC

Thermal shock (TS)

Test results of thermal shock under 2 different conditions, i.e. -55°C to 125°C and -65°C to 150°C, are summarized in Figure 17. With both conditions, test packages with solder voids (built using paste process) had more failures earlier compared to those with minimal voids that were built with ball/flux process. With milder condition of -55°C to 125°C, there was no failure even up to 2000 cycles with test packages built with ball/flux process. Whereas with condition of -65°C to 150°C, all the test packages with higher void % failed from 500+ cycles onwards, compared to the minimal void test packages which only started to fail around 1500 cycles.

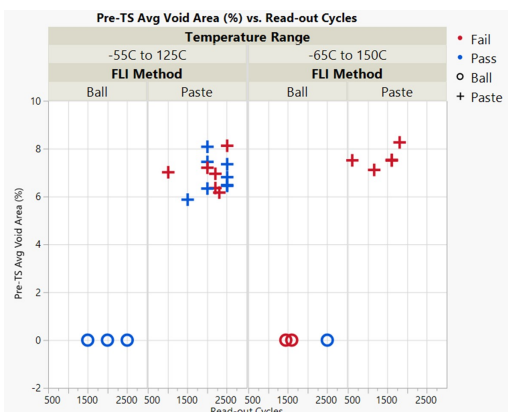


FIGURE 17. PRE-TS AVERAGE VOID VS TS FAILURE CYCLE COUNT

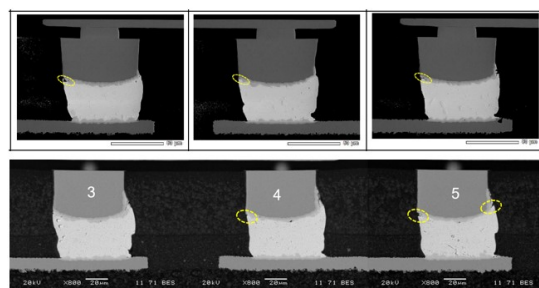
Top SEM images on Figure 18 shows the cross-sections of the test packages built with ball/flux (minimal void) process which passed 2000 cycles of TS, the joints only had some micro-cracks starting to form at the Cu pillar and solder interface. The bottom pictures are failed test packages built using paste process which resulted in some voids. These had more defined cracks that spanned across the Cu pillar-solder interface, solder-pad interface as well as some in the middle of the joint. Some micro-voids were still observed within the solder joints.

SUMMARY

Test data from all three reliability tests showed that test packages with micro solder voids performed worse when compared to test packages with minimal solder voids. The impact of voids was more apparent in electromigration, where the presence of voids reduced the mean time to failure by more than 50%. Micro voids accelerated electromigration. It is also confirmed that if the c-axis [001] of the β -Sn grain is oriented in the current direction, large diffusion of β -Sn

and formation of IMCs will affect the reliability performance of the solder joints. For thermal cycle and thermal shock, more crack propagation was observed along the different interfaces in higher void test packages.

(A) Ball Sample after 2000 cycles TS



(B) Failed Paste Sample after TS

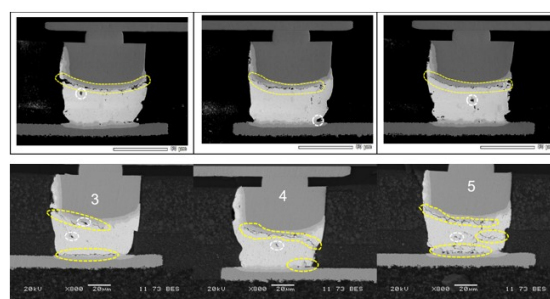


FIGURE 18. CROSS-SECTION OF PASS AND FAILED TS SAMPLE
(NOTE: YELLOW CIRCLES HIGHLIGHT THE CRACK, WHITE CIRCLES
HIGHLIGHT VOID)

ACKNOWLEDGEMENT

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